



United International University

B.Sc. in Computer Science and Engineering (CSE)

CSE 4891: Data Mining

Midterm Exam : Spring 2026 Time: 1 Hour 30 Minutes Marks: 30

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Answer all of the following questions.

1. How do you deal with imbalanced datasets where certain classes are underrepresented? Perform the stratified k-fold cross-validation on the following dataset, where $k=3$. [6]

ID	Temperature	Hungry	Air quality	Play
D1	Low	No	Fair	No
D2	Low	Yes	Excellent	No
D3	Medium	Yes	Fair	Yes
D4	High	No	Fair	Yes
D5	High	Yes	Fair	Yes
D6	High	Yes	Fair	No
D7	Low	Yes	Excellent	Yes
D8	Medium	No	Excellent	No
D9	Medium	Yes	Fair	Yes
D10	High	No	Excellent	No

2. A research team has developed a multi-class classification model with three classes: C1, C2, and C3. After justification, the following confusion matrix was obtained: [8]

Classes	Actual classes			
	----	C1	C2	C3
Predicted classes	C1	40	20	5
	C2	8	51	22
	C3	6	25	45

Estimate specificity. Explain how you can draw the 'ROC' curve for a multi-class classification problem.

3. We have a sales Cube with 3 dimensions as follows: [8]
 Product $P=\{\text{Knapsack, Suitcase}\}$; Region $R=\{\text{Dhaka, Sylhet}\}$; Time $T=\{Q1, Q2\}$
 We define the sales cube as a function:

$$\text{Sales: } P \times R \times T \rightarrow \mathbb{R}$$

The recorded values are as follows:

Sales(Knapsack, Dhaka, Q1)=6020;	Sales(Knapsack, Sylhet, Q1)=7030
Sales(Suitcase, Dhaka, Q1)=4070;	Sales(Suitcase, Sylhet, Q1)=5080
Sales(Knapsack, Dhaka, Q2)=6510;	Sales(Knapsack, Sylhet, Q2)=7290
Sales(Suitcase, Dhaka, Q2)=4560;	Sales(Suitcase, Sylhet, Q2)=5240

Using this Cube, answer the following:

- ROLL-UP: Compute the total sales per product and time across all regions.
 - SLICE: Perform a slice operation to extract the sales cube for **Q1** only.
 - DICE: Perform a dice operation to select the sub-cube for Product = {Knapsack} and Region = {Sylhet} across all quarters.
4. Consider the following dataset related to playing conditions: [3+1+2+1+1]

Day	Outlook	Humidity	Wind	Play
1	sunny	high	weak	no
2	sunny	high	strong	no
3	overcast	high	weak	yes
4	rain	high	weak	yes
5	rain	normal	weak	yes
6	rain	normal	strong	no
7	overcast	normal	strong	yes
8	sunny	high	weak	no
9	sunny	normal	weak	yes
10	rain	normal	weak	yes

Using the dataset above, answer the following:

- Calculate the **entropy** of the dataset and the **information gain** for each attribute (**Outlook**, **Humidity**, and **Wind**) at the root node.
- Identify which attribute should be selected as the **root node**, showing all necessary calculations.
- Continue the process recursively to build the complete decision tree.
- Draw the final decision tree clearly.
- Using your constructed tree, predict whether the game will be played (**Play = Yes/No**) for the following conditions:

Outlook = Sunny, Humidity = High, Wind = Strong